

Percent Applications

Finding the percent rate, finding the whole, and percent increase or decrease

Directions: Every problem below gives you two of the three numbers in the same relationship and asks for the third. Set up the proportion first, then solve. Show your work; round only where the problem says to.

The pattern to copy

$$\frac{\text{part}}{\text{whole}} = \frac{\text{percent}}{100}$$

Worked model (rate). 27 of 45 tickets were sold. What percent were sold?

$$\frac{27}{45} = \frac{p}{100} \Rightarrow 45p = 2700 \Rightarrow p = 60, \text{ so } 60\% \text{ were sold.}$$

Worked model (whole). 27 tickets is 45% of the printed run. How many were printed?

$$\frac{27}{n} = \frac{45}{100} \Rightarrow 45n = 2700 \Rightarrow n = 60, \text{ so } 60 \text{ were printed.}$$

Same three numbers, different unknown — read carefully which one is missing.

1. In one section of a theater, 18 of the 60 seats are filled. What **percent** of the seats are filled?

Answer: _____

2. Dev answered 21 of the 25 questions on a quiz correctly. What **percent** did he answer correctly?

Answer: _____

3. Exactly 40% of the hiking club walked to the trailhead. That was 12 members. **How many members** are in the club?

Answer: _____

4. A soccer player made 36 of her 45 penalty attempts. What **percent** did she make?

Answer: _____

5. A jacket was marked down by 18 dollars, and that discount was 15% of the original price. What was the **original price**, in dollars?

Answer: _____

6. Of the 250 people surveyed, 45 chose blue as their favorite color. What **percent** chose blue?

Answer: _____

7. A quality inspector found 6 cracked tiles in a shipment, which was 3% of the shipment. **How many tiles** were in the shipment?

Answer: _____

8. A bakery's weekend sales rose from 250 loaves to 300 loaves. By what **percent** did sales increase?

Answer: _____

9. A purchase of 80 dollars came with 5.40 dollars of sales tax. What is the **tax rate**, as a percent?

Answer: _____

10. In a school election, 155 students voted yes, and that was 62% of everyone who voted. **How many students** voted?

Answer: _____

11. A bike that used to cost 96 dollars now costs 72 dollars. By what **percent** did the price decrease?

Answer: _____

12. A gym's membership grew by 20% this year and now stands at 168 members. **How many members** did the gym have before the increase?

(Careful: 168 is not the part that was added — it is the new whole, which is 120% of last year's membership.)

Answer: _____